

Stochastic planning of energy system transformation pathways under uncertain industry demands

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Abstract—The transition of industrial sectors to achieve climate neutrality is imperative for effectively meeting the energy and climate policy objectives. Transforming these diverse sectors faces uncertainties, resulting in diverging end-use demands and effects on optimal transformation pathways. We use a scalable stochastic planning framework for optimising robust transformation pathways in a two- and multi-stage setup. By analysing changes in recourse decisions, including shifts in technology investments and modifications in system operation, such as imports of renewable hydrogen, we enhance decision support when facing the considered industry sector uncertainties. Our findings indicate that robust investments lead to higher lifetime system costs. Industrial demand variations affect electrolyser and storage capacities more than other capacities in the system. Moreover, weather uncertainty has a larger impact on system costs than industry uncertainties.

Index Terms—Multistage Stochastic Linear Programming, Integrated Energy System Planning, Industry demand, Uncertainty, Climate Neutrality

I. INTRODUCTION

Industrial sectors assume a vital role in the transformation to climate-neutral energy systems. Redesigning industrial processes by introducing new technologies and alternative production processes involves diverse reconfigurations of entire process chains and production sites. Moreover, these complex transformations unfold against multiple uncertainties, including economic fluctuations, technological advancements, shifting consumer preferences, future regulation, and strategic behaviour of actors, to name only a few. From an integrated energy system planning perspective, it is therefore challenging to predict future demands for existing and new energy carriers, see corresponding ranges in [1, 2, 3, 4]. For instance, due to different industry developments in Europe, the results in [1] show expected annual hydrogen demands ranging from 415 to 1785 TWh/yr, underscoring the high level of uncertainty. Addressing this uncertainty when identifying transformation pathways requires new planning tools and methods, as existing approaches typically rely on static deterministic scenarios without considering uncertainty. Understanding the uncertainty impacts of electricity and hydrogen consumption in the industry sector becomes crucial for the national production and storage of hydrogen and for imports from outside Europe.

The remainder is organised as follows. Section II provides an overview of relevant previous work. Section III explains the applied methodology involving the stochastic EMPRISE

framework for planning integrated energy systems under uncertainty. Section IV sets up the structure and assumptions of the case study. Section V presents and discusses the optimisation model results. Section VI closes with a summary and draws relevant conclusions.

II. PREVIOUS WORK

In recent years, there has been an uptick in research focusing on the decarbonisation of industry sectors, signalling a growing recognition of its critical importance in achieving net-zero energy systems through innovative approaches, policy frameworks, and technological breakthroughs.

A study by the European Commission [1] shows possible pathways for the decarbonisation of the European industry sector, the resulting energy demands and the impact on the overall energy system. The effect of decarbonising the industry on European power and hydrogen systems using an iron and steel industry example is analysed in [5] using the model METIS. The authors in [6] describe carbon capture developments and hydrogen uptake in the power, heat, and industry sectors towards 2060 without Russian gas imports. Case studies for the industry transformation from fossil fuels to hydrogen, focusing on Germany, are given in [7]. There are also many publications on the decarbonisation of individual industries, e.g. the construction industry in [8], the oil refining industry in [9], and the cement industry in [10].

An exploration of how varying developments in industry demands affect outcomes through scenario-based sensitivity analysis is illustrated in [1]. However, the literature reveals a gap in frameworks that integrate these demands with an endogenous treatment of the inherent uncertainties. Incorporating endogenous uncertainty within integrated energy system models remains scarce, with notable exceptions such as [11], which utilises a multi-horizon stochastic programming method to navigate decision-making under uncertainty in the power sector. Although alternative methodologies for detailed integrated energy system planning are known by, e.g. [12, 13], they primarily employ scenario-based sensitivity analysis to expose uncertainty. Recent work in [14] quantifies uncertainties in multi-period energy system planning with endogenous consideration of decommissioning decisions and market uptake restrictions. However, these approaches do not make decisions under uncertainty, and we address this gap with the contributions listed below.

- Consideration of different European industry developments as a parametric uncertainty in an integrated energy system model,
- Quantification of the effects with a two- and multi-stage stochastic optimisation framework,
- Comparison of a two-stage and a multi-stage scenario tree configuration.

III. METHODOLOGY

A. Conceptual approach

The modelling and optimisation framework EMPRISE for integrated energy system transformation pathways solves stochastic multi-period capacity expansion and system operation problems, for details, see [15]. It is formulated as a multi-stage linear stochastic program and it can handle different decision and scenario tree structures when building its instances. These determine how often uncertainty is revealed over the decision stages and which variables are the here-and-now or the wait-and-see decisions, see Figure 3.

The EMPRISE framework is implemented in Python and uses the packages `Pyomo` [16, 17] and `mpi-sppy` [18]. It contains generic modelling components for representing the integration of energy and non-energy commodities and their underlying sectors, markets, infrastructures, and technologies. A schematic illustration of the modelling components, e.g. nodes, commodities, sources, sinks, storage systems, links, and converters, and their important interactions is shown in ???. A complete list of all case study specific components can be found in the appendix.

Analysing system transformation in development trajectories requires modelling different temporal dimensions and dependencies. For this purpose, the EMPRISE framework distinguishes between two temporal concepts: planning periods and time slices with individual time steps (full-year or representative plus critical time slices). The connection between individual slices and the planning periods results in the desired closed dynamic multi-period planning. The operation of the integrated energy system is modelled in hourly resolution. The analysis of the system transformation requires the representation of several planning periods in the model.

B. Scenario-specific optimisation problem

The modelling framework's objective is to minimise the expected total system cost of the energy system transformation pathway, incorporating investment costs of multi-period capacity expansion planning and operational costs of supplying end-use demands. The objective function of each scenario tree branch accounts for the costs of investment ("INV") and system operation ("SYSOP") decisions. The resulting scenario-specific linear programming problem writes as

$$\min f_{\xi}(\mathbf{x}_{\xi}) = \sum_{p \in P} \mu_p \left(f_{\xi,p}^{\text{INV}}(\mathbf{x}_{\xi}) + \sum_{t \in T} f_{\xi,p,t}^{\text{SYSOP}}(\mathbf{x}_{\xi}) \right) \quad (1)$$

s.t. Capacity restrictions, Expansion continuity,

Dispatch characteristics, Nodal commodity clearing,

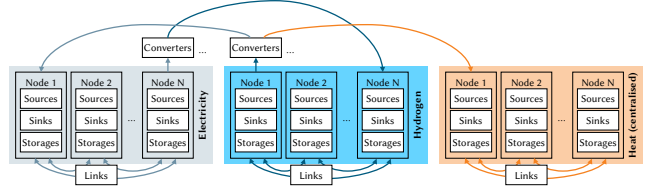


Figure 1: Schematic illustration of the EMPRISE components and their interaction for three exemplary commodities.

where $\mu_p \in \mathbf{R}$ is a period-specific discount factor to account for the time value of money and perpetuity in the multi-period planning problem, and $\mathbf{x}_{\xi}$ is the vector notation of all scenario-specific decision variables. For ease of formulation, we denote the set of feasible solutions of a scenario $\xi \in \Xi$ as $\mathcal{X}_{\xi}$.

C. Multi- and two-stage stochastic problem

The multi-stage stochastic linear problem minimising the expected costs of the multi-period capacity expansion planning problem of the EMPRISE framework can be written as:

$$\min \sum_{\xi \in \Xi} \pi_{\xi} f_{\xi}(\mathbf{x}_{\xi}) \quad \text{s.t.} \quad \mathbf{x}_{\xi} \in \mathcal{X}_{\xi}, \mathbf{x}_{\xi} \in \mathcal{X}^{\text{na}}, \quad (2)$$

where $\pi_{\xi} \in [0, 1]$ denotes the scenario-specific probability, and $\mathcal{X}^{\text{na}}$ denotes the feasibility set required for enforcing the nonanticipativity constraints of the underlying scenario tree structure. The nonanticipativity constraints ensure that all capacity expansion variables of a scenario ξ are consistent with their split copies in all indistinguishable scenarios $\{\xi\}$ of scenario ξ .

The EMPRISE framework can consider strategic and operational uncertainties that can be implemented through scenario-specific parameters. Here, we focus on the operational uncertainties, which capture the complex correlations between the system components' time-series signals based on historical data and the uncertain industry demands.

D. Decomposition and solution approach

To solve large model instances, we use a decomposition and solution approach, that is based on the Progressive Hedging Algorithm (PHA) [19], which is a scenario-based decomposition algorithm for large-scale stochastic optimisation problems [20]. By relaxing the nonanticipativity constraints and decomposing the explicit (or extensive form) formulation of the multi-stage stochastic problem, the distributed algorithm solves individual scenario subproblems in an iterative manner and penalises the nonanticipativity violations.

While the main processes perform the progressive hedging algorithms and generate candidate solutions, additional processes compute lower and upper bounds based on the candidate solutions. The lower bounds are obtained from the procedure described in [20], which uses dual prices that are calculated during the execution of the standard PHA. Upper bounds, the best-known solution to the primal problem from incumbent solutions, are calculated by shuffling the scenarios and looping over them to try a candidate solution for the nonanticipative variables. Since the investment variables

are fixed, the resulting scenario-specific subproblems become economic dispatch problems of the underlying energy system, which are much easier to solve.

In this case study, the subproblems have been solved with the Barrier (interior point) algorithm of Gurobi Optimizer Version 11.0.0 [21] on a medium-range HPCC node. For the two-stage cases, nine tasks on 18 CPUs were executed at the same time, and, for the multi-stage cases, 81 tasks on 162 CPUs were executed at the same time.

IV. CASE STUDY DESCRIPTION

We use EMPRISE to examine different potential developments in Europe's industry sectors. The analysis adopts a two- and multi-stage stochastic approach, considering investment ("INV") and system operation ("SYSOP") decisions for energy system components in the planning periods 2025, 2035, and 2045. The scenario tree considers investment decisions nonanticipative, i.e. the decisions describing the system developments must be robust against all future revelations of uncertainty in the later decision stages.

A. Industry demand developments in Europe

To investigate the European industry sector uncertainty, we encode three different industry electricity and hydrogen end-use demand scenarios into the scenario trees. The assumptions for industry demand development are taken from [1] and defined as follows:

- Scenario Electricity (*sc_El*): Focus on electrifying industry processes in the decarbonisation process and a higher hydrogen import price to encourage higher electrification rates in the heating sector transformation.
- Scenario Hydrogen (*sc_H2*): Focus on including more hydrogen in the decarbonisation of the industry sector and more hydrogen in the heating sector through lower hydrogen import prices.
- Scenario Relocation (*sc_Re*): Focus on relocating industry processes to non-European countries with new global value chains in decarbonisation and a higher hydrogen import price to encourage higher electrification rates in the heating sector.

The end-use demands for Europe and the import price assumptions for green hydrogen are summarised in Figure 2. The trajectories exhibit a large increase in hydrogen demand and a slight increase in electricity demand for the scenarios *sc_El* and *sc_H2* starting in 2030. A slow increase in hydrogen demand characterises the *sc_Re* scenario, and electricity demand will almost stagnate until 2050.

B. Scenario tree configuration

We combine the three identified industry scenarios in a two-stage scenario tree, which allows for robust investment decisions in the pathway. Figure 3 (top) shows this two-stage configuration. We choose exogenous probabilities of the single scenarios for the base case as follows: scenarios *sc_El* and *sc_H2* have a probability of 0.4 each, *sc_Re* has a probability of 0.2, because *sc_Re* was considered less likely in this case

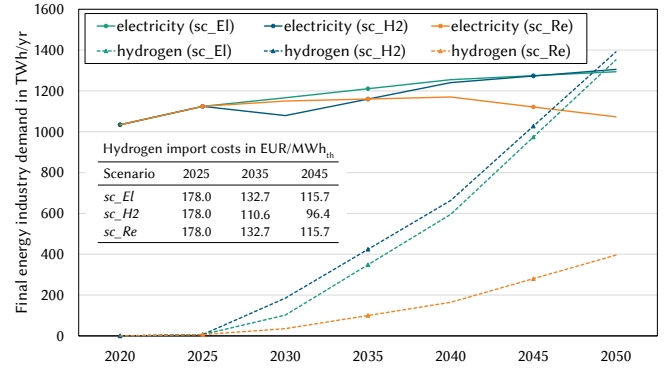


Figure 2: Final European energy industry demand development for three different scenarios (2020 to 2050) (without electrolysis, space cooling & heating, steam & hot water) and hydrogen import prices, based on the results in [2].

study setup. In the two-stage case, all investment decisions are taken in the first stage, and all system operation decisions are in the second stage, see Figure 3 (top).

Figure 3 (bottom) also shows the second case, which represents the multi-stage setup. The probabilities for the three industry scenarios remain the same, but the branches are now split up to $3^3 = 27$ individual transformation pathways as every possible combination of industry development is considered. In the multi-stage case, the investment decisions in 2025 are robust against all uncertainties in the future. In Figure 3 (bottom), three and nine individual investment decisions can be made for 2035 and 2045, respectively, and the conditional probability in stage four is the product of the three individual probabilities of the scenario path.

To evaluate the robustness and sensitivity of the stochastic solutions, the probability of different future industry developments is varied to assess their impacts on total system costs. Moreover, the analysis incorporates different historical meteorological years to showcase the dependency on other uncertainties. We use the reference case of 2012 featuring a cold "Dunkelflaute" event in early February.

Note that further input data on costs, existing capacities and parameter are available in an open-access repository at [10.5281/zenodo.11186578](https://zenodo.org/record/11186578) [22].

V. RESULTS AND DISCUSSION

A. Convergence results

The first results show the convergence behaviour of the EMPRISE model instances for the two- and multi-stage problems. Figure 4 shows the lower and upper bounds obtained during the PHA iterations. Strong duality holds since the EMPRISE framework results in convex linear and quadratic programming problems. The initial solutions of the subproblems differ more in the multi-stage case and the final solution of the two-stage approach results in higher expected total costs.

Figure 4 also illustrates the corresponding relative gap trajectory over the PHA iterations. The trajectories show that convergence below 1% is reached after five iterations in the two-stage case and after two iterations in the multi-stage case.

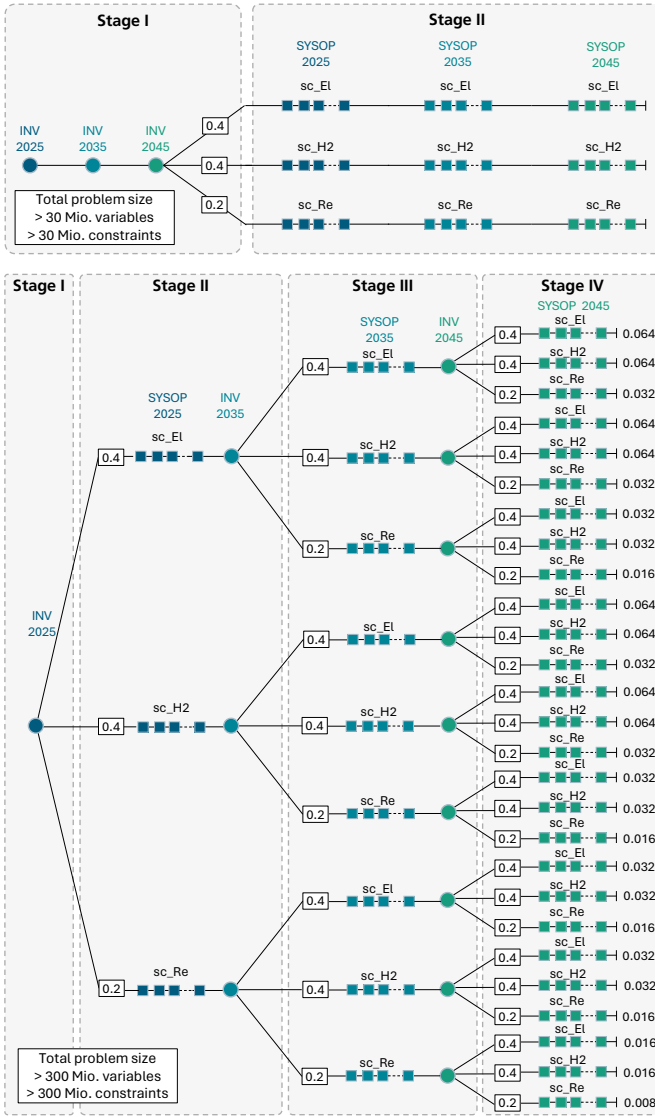


Figure 3: Scenario tree for the two-stage (top) and the multi-stage (bottom) setting of the case study.

Convergence below 0.1 % is reached after 16 iterations in the two-stage and after 14 iterations in the multi-stage case. Faster convergence and lower objective values for the multi-stage case may be explained by fewer nonanticipativity variables and smaller conditional probabilities for each of the 27 scenarios.

B. Objective value results

Table I shows the resulting objective function values for different scenario configurations, i.e. the resulting total lifetime costs of the energy system. The relative deviation always refers to the variant with the highest costs, i.e. sc_EI for 2012.

As expected, the lifetime cost differences between sc_H2 and sc_EI are limited. sc_Re results in lower costs due to its lower exogenous demand in the industry sector. Note that this is only related to the energy system costs that we consider in the model, not to Europe's total economic system.

Equal probability weights of all three scenarios result in lower total costs than the weighting of 0.4/0.4/0.2 since

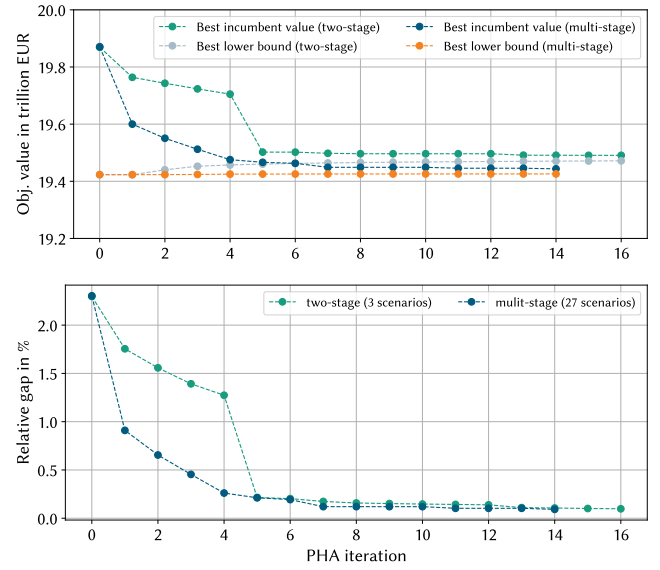


Figure 4: Lower and upper bound (top) and relative gap (bottom) trajectories over the PHA iterations for the multi- and the two-stage approach, where the objective value corresponds to total discounted lifetime costs of investment and system operation decisions.

Table I: Objective function values and deviations for the industry scenarios with different stochastic scenario tree setups.

Meteo. year scenario tree	Probability of $[sc_EI, sc_H2, sc_Re]$	Obj. value in tn. EUR	Relative deviation
2012 single	[1.0, -, -]	20.11	-
2012 single	[-, 1.0, -]	19.89	-1.13 %
2012 single	[-, -, 1.0]	17.12	-17.50 %
2012 multi-stage	[0.4, 0.4, 0.2]	19.45	-3.41 %
2012 two-stage	[0.4, 0.4, 0.2]	19.49	-3.19 %
2012 multi-stage	[0.34, 0.33, 0.33]	19.13	5.16 %
2012 two-stage	[0.34, 0.33, 0.33]	19.20	-4.73 %

the most favourable scenario has higher objective function weights. The probability of reduced industry demands increases compared to higher demand scenarios. Moreover, the multi-stage setup shows slightly lower total costs than the two-stage setup. In the latter, all pathway investment decisions have to align, whereas in the multi-stage case, only the bundled investment decisions have to be aligned, i.e. nonanticipative.

The influence of the meteorological year as an operational uncertainty can be seen in Table III. Using a two-stage scenario tree for equiprobable meteorological years 2006 to 2012 in combination with one of the industry scenarios shows that the objective function value is lower than using only 2012 as the reference year because 2012 is the most expensive weather year. In this two-stage setting, it has only a probability of $\frac{1}{7}$. The combination of 2010 and 2012 and the three scenarios results in slightly higher costs than using only 2012. These examples demonstrate the pivotal role of the meteorological year in shaping investment and operation decisions in the energy system. Its inherent uncertainty underscores its role in reliable energy system planning.

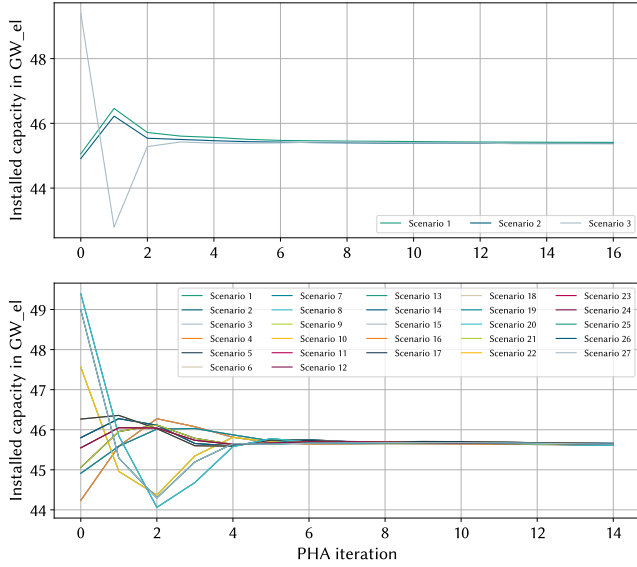


Figure 5: Candidate investment decisions over the PHA iterations for gas turbines in Germany for 2025 in the two-stage (top) and multi-stage (bottom) setups, own illustration.

C. Convergence of robust decision-making

The information collected during the PHA iterations exposes the convergence trajectory of the nonanticipative decisions. For each iteration and scenario of the two- and multi-stage setups, Figure 5 shows the candidate gas turbine investment decisions in Germany for the planning period 2025. The results show that the initially heterogeneous capacities of the first iterations converge over the PHA iterations, and re-establish the nonanticipativity conditions relaxed by the scenario-decomposition approach. While the range of candidate capacities after the first iteration varies between 49.4 GW and 45.0 GW in the two-stage and between 49.4 GW and 44.2 GW in the multi-stage case, the value for all scenarios after the PHA iterations is 45.4 GW in the two-stage and 45.6 GW in the multi-stage case. The probability-weighted mean value of the initial investment decision in the two-stage case is 45.9 GW, which deviates more than 0.5 GW from the result gained from the stochastic solution. Figure 6 shows the investment decision for electrolyser capacity in Germany and the convergence over the PHA iterations in 2045. The initial capacities range from 45.5 GW to 18.7 GW and the converged capacities in the multi-stage case from 45.0 GW to 20.2 GW, which are bundled to nine different investment decisions. In the two-stage case the capacities result in 35.4 GW.

D. Capacity expansion and hydrogen import results

The last planning period (2045) can be seen as a steady-state system in a multi-period planning approach. Table II shows the results of the three scenarios, their weighted mean value, and the solution of the two-stage stochastic optimisation problem.

The difference between the weighted mean and the two-stage results for generator capacities is slight for thermal and renewable power plants. Still, the observed differences for

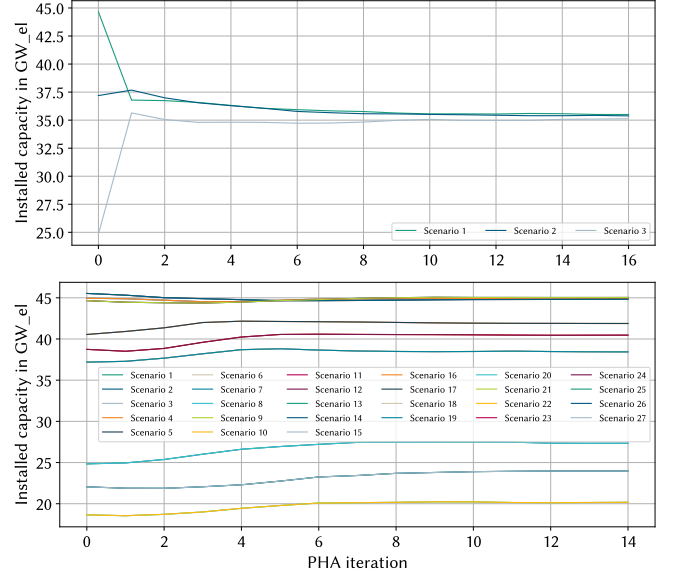


Figure 6: Candidate investment decisions over the PHA iterations for electrolyzers in Germany for 2045 in the two-stage (top) and multi-stage (bottom) setup, own illustration.

Table II: Capacity investment decisions and renewable hydrogen imports for Europe in 2045.

Capacity in GW _{el}	<i>sc_El</i>	<i>sc_H2</i>	<i>sc_Re</i>	weighted mean	two-stage
OCGT / CCGT	259.7	260.3	268.9	261.8	261.7
CCGT w. CHP	49.9	47.1	46.7	48.1	47.9
Onshore wind	744.8	730.5	709.1	732.0	728.4
Offshore wind	214.5	200.9	173.1	200.8	203.7
Solar PV	2 310.3	2 299.0	2 238.9	2 291.5	2 297.4
Electrolyser	184.8	148.5	98.3	153.0	146.5
Li-Ion battery	319.5	340.2	352.9	334.5	346.4
H ₂ import in TWh _{th} /yr	589	712	81	-	658/707/7

electrolyzers and battery storage are more prominent between the stochastic two-stage results and the weighted mean of the three deterministic scenarios. More than generator capacity, industrial demand variations affect flexibility capacities, such as storage and electrolyzers. Hydrogen imports are mainly dependent on hydrogen demand and the import price.

VI. SUMMARY AND CONCLUSION

Our findings explore and quantify the relationship between the probabilities of different industry developments and the expected costs and system configurations associated with transforming the energy system. We analyse changes in resource decisions, including technological investment shifts and system operation modifications, such as renewable hydrogen imports. By highlighting more robust transformation pathways towards a climate-neutral system, we demonstrate how we can use a new modelling and optimisation framework, EMPRISE, to enhance decision support in the face of industry sector uncertainty. Further investigations must look into differences between two- and multi-stage stochastic optimisation.

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APPENDIX

Additional information on the case study setup of the EMPRISE instances:

- **Nodes** – 21 European countries: Austria, Belgium, Switzerland, Czech Republic, Germany, Denmark, Spain, Finland, France, United Kingdom, Hungary, Ireland, Italy, Luxembourg, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, and Sweden.
- **Converters** – Thermal power plants (OCGT, CCGT, district heating CHP, industry CHP, nuclear, coal, lignite, oil), renewable generators (onshore Wind, offshore Wind, rooftop solar PV, utility scale solar PV), electrolyser, methane reactor, direct air capture, air and ground heat pumps, direct restive heating units, boiler, electric vehicle, air conditioners.
- **Commodities** – electricity markets, hydrogen markets, methane markets, carbon dioxide markets, carbon dioxide budget, industrial central heat markets, district heating markets, decentralised building heating, decentralised industry heating, road traffic.
- **Sources** – Hydropower, biomass, natural gas import, green hydrogen import, emergency power generation, onshore and offshore wind power, solar PV, nuclear, oil, lignite, coal.
- **Storages** – Battery storage, natural gas cavern, hydrogen tanks, carbon dioxide storage, cold storage, district heating storage, industry heating storage, residential heat storage.
- **Links** – Cross-border interconnections: High-voltage three-phase current transmission (HVAC) and High-voltage direct current transmission (HVDC) of the transmission grids (onshore and offshore), natural gas, hydrogen and CO₂ pipelines.
- **Sinks** – End-use demands for electricity, methane, hydrogen, industry heating, residential heating, cooling, passenger road transport, heavy-duty road transport.

Table III: Objective function values and deviations for investigating the three industry scenarios with different stochastic scenario tree setups and meteorological years.

Meteo. year and scenario tree	Probability of scenario [<i>sc_El</i> , <i>sc_H2</i> , <i>sc_Re</i>]	Probability of meteorological year ['12, '11, '10, '09, '08, '07, '06]	Obj. value in tn. EUR	Relative deviation
2012 single	[1.0, -, -]	[1.0, -, -, -, -, -]	20.11	-
2012 single	[-, 1.0, -]	[1.0, -, -, -, -, -]	19.89	-1.13%
2012 single	[-, -, 1.0]	[1.0, -, -, -, -, -]	17.12	-17.50%
2012 multi-stage	[0.4, 0.4, 0.2]	[1.0, -, -, -, -, -]	19.45	-3.41%
2012 two-stage	[0.4, 0.4, 0.2]	[1.0, -, -, -, -, -]	19.49	-3.19%
2012 multi-stage	[0.34, 0.33, 0.33]	[1.0, -, -, -, -, -]	19.13	-5.16%
2012 two-stage	[0.34, 0.33, 0.33]	[1.0, -, -, -, -, -]	19.20	-4.73%
2010 single	[1.0, -, -]	[-, -, 1.0, -, -, -]	18.97	-5.99%
2010 single	[-, 1.0, -]	[-, -, 1.0, -, -, -]	18.74	-7.33%
2010 single	[-, -, 1.0]	[-, -, 1.0, -, -, -]	15.95	-26.12%
2010/12 two-stage	[0.4, 0.4, 0.2]	[0.5, -, 0.5, -, -, -]	19.74	-1.89%
2006-12 two-stage	[1.0, -, -]	[1/7, 1/7, 1/7, 1/7, 1/7, 1/7]	19.31	-4.14%
2006-12 two-stage	[-, 1.0, -]	[1/7, 1/7, 1/7, 1/7, 1/7, 1/7]	19.47	-3.32%
2006-12 two-stage	[-, -, 1.0]	[1/7, 1/7, 1/7, 1/7, 1/7, 1/7]	16.82	-19.54%